CHAPTER 8

Quine–McCluskey (Tabulation) Method

A step-by-step table method · prime implicants · essential prime implicants · don't-cares

QUICK TRICK
Good news: this chapter is short. It is the K-map idea (join neighbours) written as a **table**. If you understood Chapter 7, you already know the idea.

8.1 Why do we need another method?

● **LEVEL 1 · BEGINNER**

K-map has a limit. It is easy for 2, 3 or 4 variables. For 5 or 6 variables it becomes a big, confusing picture. For 10 variables it is impossible.

The Quine–McCluskey (QM) method does the same job with **tables**. A computer can follow it step by step, so it works for any number of variables. It is also called the **tabulation method**.

Analogy. Sorting socks. First put them in piles by colour, then match pairs, then match pairs of pairs. QM is exactly this: make piles, match neighbours, repeat.

	K-map	Quine–McCluskey
Style	Picture (look and circle)	Table (step by step)
Best for	2 to 4 variables (5 with effort)	5 or more variables, computers
Risk	Missing a wrap-around group	Making a small mistake in a long table
Speed in the exam	Fast	Slow (use only if asked)

8.2 The basic idea

● **LEVEL 2 · EXAM**

Two minterms can be joined if they differ in **exactly one bit**. The bit that is different becomes a **dash (–)**. A dash means “this variable disappeared”.

```
0110  and  0100   differ only in the 3rd bit   ->   01-0    (that is A'BD')
1011  and  1111   differ only in the 2nd bit   ->   1-11    (that is ACD)
```

QUICK TRICK
Pattern → term: 1 → plain letter, 0 → letter with a bar, dash → skip the letter. Example (A B C D): **1-01** → A · C' · D = **AC'D**.

QUICK TRICK
Dash count trick: a pattern with **k dashes** covers 2^k **minterms**. 01-0 (1 dash) covers 2 minterms. --01 (2 dashes) covers 4 minterms. Use this to check your table.

Words: a joined term that cannot be joined any more is a **prime implicant (PI)**. A PI that is the only one covering some minterm is an **essential prime implicant (EPI)**. (Same meaning as in the K-map.)

8.3 The steps

```
STEP 1  Write all minterms (and don't-cares) in binary.
        Put them in piles by the NUMBER OF 1s.   (Column 1)
STEP 2  Compare each pile only with the NEXT pile.
        If two terms differ in exactly one bit, join them (put a dash there).
        Tick (v) both terms. Write the new term in the next column.
STEP 3  Repeat on the new column. Dashes must be in the SAME place to join.
        Stop when nothing can be joined.
STEP 4  Terms with NO tick are the PRIME IMPLICANTS.
STEP 5  Draw the PRIME IMPLICANT CHART (rows = PIs, columns = minterms).
STEP 6  A column with only ONE mark gives an ESSENTIAL PI. Take it.
STEP 7  Cover any remaining minterms with as few extra PIs as possible.
STEP 8  Write the answer: OR of the chosen terms.
```

QUICK TRICK

Only compare neighbouring piles. Two terms that differ in one bit must have a different number of 1s, and the numbers differ by exactly 1. So pile 0 is compared with pile 1, pile 1 with pile 2, and so on. Never compare pile 0 with pile 2.

QUICK TRICK

Quick "one-bit" test: the bigger term must look like the smaller one **plus one extra 1**. 0101 (5) and 0111 (7): yes. 0011 (3) and 0101 (5): no, two bits changed.

8.4 Worked examples

Example 1 — three variables · Level 1

F(A,B,C) = Σm(0, 1, 3, 7)

Column 1

1s	Minterms	Pattern	
0	0	000	✓
1	1	001	✓
2	3	011	✓
3	7	111	✓

Column 2

1s	Minterms	Pattern	
0	0,1	00-	
1	1,3	0-1	
2	3,7	-11	

No ticks in Column 2 (nothing more can be joined). So the prime implicants are the three terms of Column 2: **00– = A'B'**, **0–1 = A'C**, **–11 = BC**.

Prime implicant chart (★ = the only mark in its column):

Prime implicant	Term	0	1	3	7
00-	A'B'	★	✓		
0-1	A'C		✓	✓	
-11	BC			✓	★

m0 has only one mark (★, row A'B') → A'B' is essential. m7 has only one mark (★, row BC) → BC is essential. Together they cover 0, 1, 3, 7. **Answer: F = A'B' + BC.**

Example 2 — four variables · Level 3 (same function as K-map Example 4)

F(A,B,C,D) = Σm(0, 1, 2, 5, 6, 7, 8, 9, 10, 14)

Column 1

1s	Minterms	Pattern	
0	0	0000	✓
1	1	0001	✓
	2	0010	✓
	8	1000	✓
2	5	0101	✓
	6	0110	✓
	9	1001	✓
	10	1010	✓
3	7	0111	✓
	14	1110	✓

Column 2

1s	Minterms	Pattern	
0	0,8	-000	✓
	0,2	00-0	✓
	0,1	000-	✓
1	1,9	-001	✓
	2,10	-010	✓
	1,5	0-01	
	2,6	0-10	✓
	8,10	10-0	✓
	8,9	100-	✓
2	6,14	-110	✓
	5,7	01-1	
	6,7	011-	
	10,14	1-10	✓

Column 3

1s	Minterms	Pattern	
0	0,2,8,10	-0-0	
	0,1,8,9	-00-	
1	2,6,10,14	--10	

Not ticked, so these are the **prime implicants**: **0-01** = A'C'D (m1,5), **01-1** = A'BD (m5,7), **011-** = A'BC (m6,7), **-00-** = B'C' (m0,1,8,9), **-0-0** = B'D' (m0,2,8,10), **--10** = CD' (m2,6,10,14).

Prime implicant chart:

Prime implicant	Term	0	1	2	5	6	7	8	9	10	14
0-01	A'C'D		✓		✓						
01-1	A'BD				✓		✓				
011-	A'BC					✓	✓				
-00-	B'C'	✓	✓					✓	★		
-0-0	B'D'	✓		✓				✓		✓	
--10	CD'			✓		✓				✓	★

Columns with a single mark: m9 (only B'C') and m14 (only CD'). So **B'C' and CD' are essential**. They cover 0, 1, 2, 6, 8, 9, 10, 14. Left over: **m5 and m7**. The PI **A'BD** (5, 7) covers both with one term (the other choices need two terms).

Answer: F = B'C' + CD' + A'BD. (The same answer as the K-map. B'D' was a PI but is not needed.)

Example 3 — with don't-cares · Level 3

F(A,B,C,D) = Σm(1, 3, 7, 11, 15) + d(0, 2, 5)

Rule: write the don't-cares (0, 2, 5) in Column 1 together with the real minterms. They help to build bigger groups.

Column 1

1s	Minterms	Pattern	
0	0	0000	✓
1	1	0001	✓
	2	0010	✓
2	3	0011	✓
	5	0101	✓
3	7	0111	✓
	11	1011	✓
4	15	1111	✓

Column 2

1s	Minterms	Pattern	
0	0,2	00-0	✓
	0,1	000-	✓
1	1,5	0-01	✓
	1,3	00-1	✓
	2,3	001-	✓
2	3,11	-011	✓
	3,7	0-11	✓
	5,7	01-1	✓
3	7,15	-111	✓
	11,15	1-11	✓

Column 3

1s	Minterms	Pattern	
0	0,1,2,3	00--	
1	1,3,5,7	0--1	
2	3,7,11,15	--11	

Prime implicants: **00--** = A'B', **0--1** = A'D, **--11** = CD.

Rule: in the chart, list only the **real minterms** (1, 3, 7, 11, 15) as columns. Don't-cares do not have to be covered.

Prime implicant	Term	1	3	7	11	15
00--	A'B'	✓	✓			
0--1	A'D	✓	✓	✓		
--11	CD		✓	✓	★	★

m11 and m15 have only one mark (CD). So **CD is essential** and it also covers 3 and 7. The only minterm left is **m1**. It can be covered by A'B' (0,1,2,3) or by A'D (1,3,5,7). Both cost the same.

Answer: F = CD + A'B' (or CD + A'D). The same as the K-map.

> **EXAM TRAP**
> Don't-cares go in **Column 1** (to build groups) but **not** in the PI chart columns (they need no cover). A PI made only of don't-cares is never chosen.

8.5 Common mistakes

> **COMMON MISTAKE**
> • Comparing piles that are not neighbours (pile 0 with pile 2).
> • Joining terms whose dashes are in different places.
> • Forgetting to tick a term that was used. A used term is not a prime implicant.
> • Writing the same new term twice. If two pairs give the same pattern, write it **once**.
> • Putting don't-cares as columns in the PI chart.
> • Reading a dash as a variable. A dash means the variable is gone.

Quick-Solve Tricks — Summary

Situation	Trick
Can two terms join?	They differ in exactly one bit, and the dashes are in the same places.
Which piles to compare?	Only neighbouring piles (number of 1s differs by 1).
Pattern → term	1 → letter, 0 → letter with bar, dash → skip.
How many minterms in a term?	k dashes cover 2^k minterms.
Which terms are prime implicants?	The ones with NO tick.
Which are essential?	A PI that has the only mark in some column of the chart.
Don't-cares	Use in Column 1. Do not list them in the chart.
Check	Compare the answer with a quick K-map for 3 or 4 variables.

GATE QUICK CHECK — Chapter 8

```
Join if the terms differ in ONE bit (same dash places).   k dashes -> 2^k minterms
Unticked terms = PRIME IMPLICANTS.    Column with one mark = ESSENTIAL PI.
Don't-cares: in Column 1, not in the chart.
```

Traps: joining non-neighbour piles · a dash mismatch · PI vs EPI · don't-cares in the chart · the answer may not be unique.

Practice questions

● **LEVEL 1 · BEGINNER**

Q1. Join 0110 and 0100. What is the result?

Q2. Can 0011 and 1100 be joined?

Q3. Write the term for the pattern 1–0– (variables A B C D).

● **LEVEL 2 · EXAM**

Q4. Put the minterms 3, 5, 6, 7 (3 variables) into piles by the number of 1s.

Q5. Find all prime implicants of Σm(0, 1, 3, 7) (3 variables).

Q6. Which of them are essential?

● **LEVEL 3 · GATE**

Q7. Find the prime implicants and the minimal SOP of Σm(2, 3, 6, 7) (3 variables).

Q8. How many prime implicants does Σm(0, 1, 2, 5, 6, 7) (3 variables) have? How many are essential?

Q9. A prime implicant in a 5-variable problem has 2 dashes. How many minterms does it cover? How many literals does the term have?

Q10. Why are don't-cares not listed as columns in the prime implicant chart?

Q11. Find the minimal SOP of F(A,B,C,D) = Σm(4, 5, 6, 7, 12, 13, 14, 15).

Q12. How many prime implicants does Σm(0, 1, 2, 5, 6, 7, 8, 9, 10, 14) (4 variables) have?

Answers

Q	Solution
1	Only the 3rd bit differs → **01–0** (A'BD').
2	**No.** They differ in all 4 bits (and their numbers of 1s are 2 and 2).
3	1 → A, 0 → C', dashes skipped. **AC'**.
4	3 = 011, 5 = 101, 6 = 110 → two 1s each (pile 2). 7 = 111 → three 1s (pile 3).
5	**A'B' (00–), A'C (0–1), BC (–11)**.
6	m0 is only in A'B' and m7 is only in BC → **A'B' and BC** are essential (A'C is not needed).
7	2 = 010, 3 = 011, 6 = 110, 7 = 111. Column 2: 01– (2,3), –10 (2,6), –11 (3,7), 11– (6,7). Column 3: –1– (2,3,6,7). One prime implicant: **B**. **F = B**.
8	**6 prime implicants** (A'B', A'C', BC', AB, AC, B'C). **None** are essential (every column has two marks).
9	2^2 = **4 minterms**. Literals = 5 − 2 = **3**.
10	Because a don't-care may be 0 or 1. The output for them does not matter, so nothing needs to cover them.
11	All terms combine into one pattern **–1––** (B = 1 in every minterm). **F = B**.
12	**6** prime implicants: B'C', B'D', CD', A'BD, A'C'D, A'BC. (See Example 2.)

Next: Chapter 9 — Combinational Circuits: half adder, full adder, subtractors, ripple carry, carry look-ahead, BCD adder, adder-subtractor.

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